

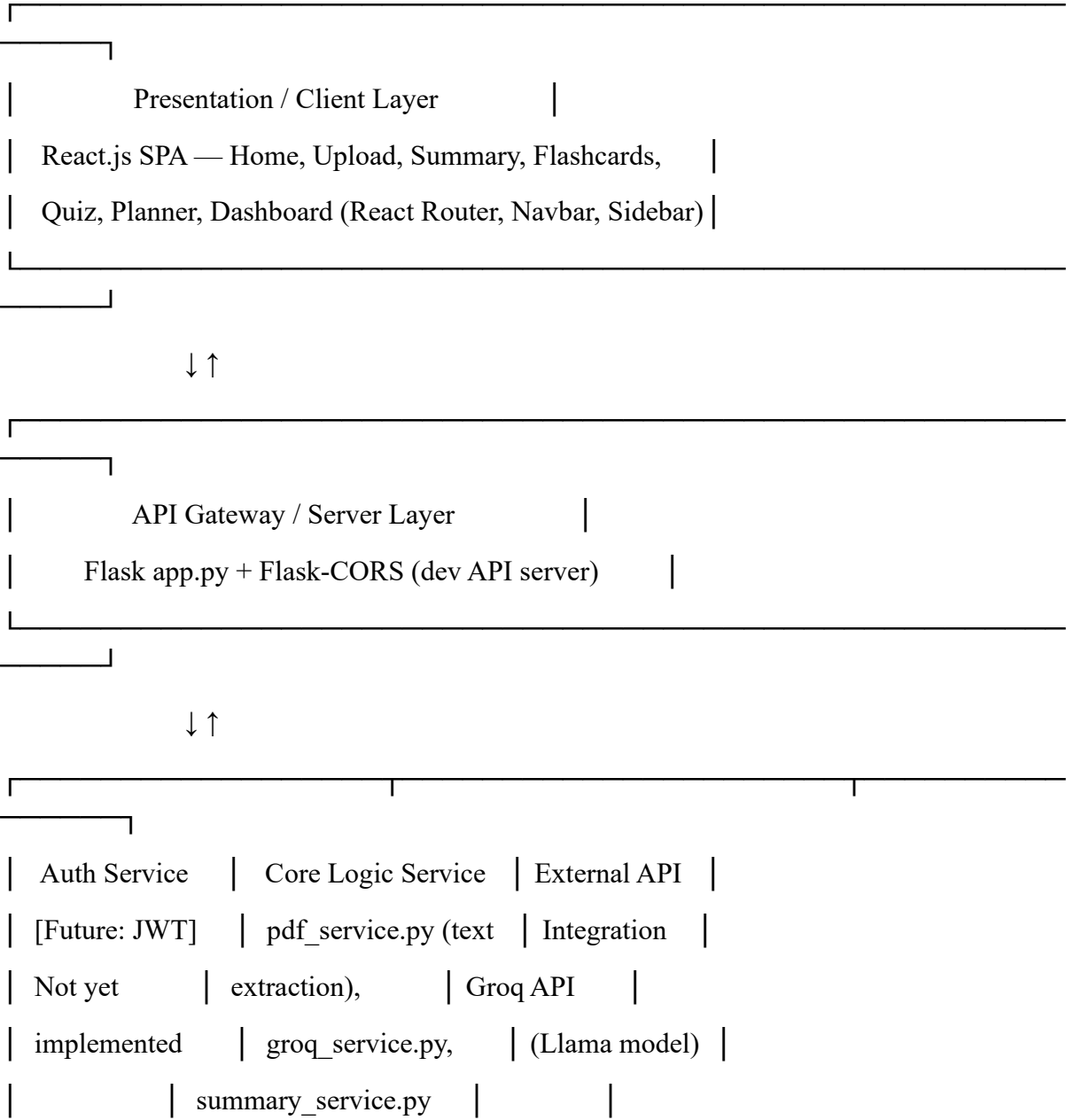
Solution Architecture

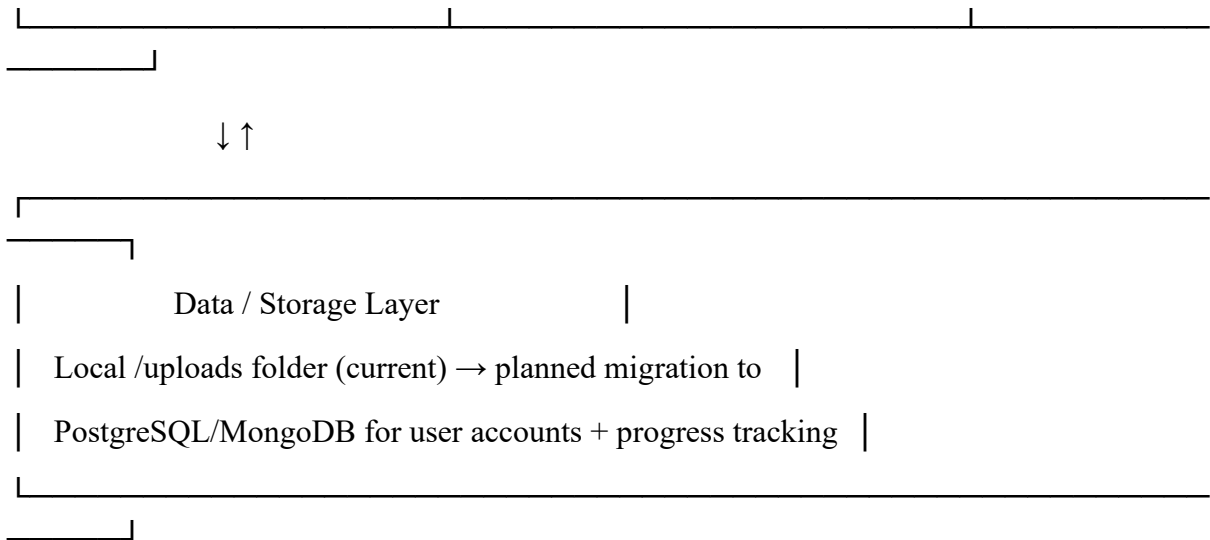
Date 06 July 2026

Team ID *****

Project Name StudyAI – AI Powered Smart Study Companion

Solution Architecture Diagram (recreate in Draw.io as a 4-layer stack):





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders Home, Upload, Summary, Flashcards, Quiz, Planner, and Dashboard pages; handles routing and calls backend via api.js	React.js, JavaScript, HTML/CSS, React Router
API Gateway	Receives frontend requests, routes to upload/summary services, handles CORS	Flask, Flask-CORS
Core Logic Service (Microservices)	Extracts text from uploaded files, sends content to Groq API, formats summaries/flashcards/quiz output	PyMuPDF (fitz), Groq API, Llama model
Database / Storage	Currently stores uploaded files locally with UUID filenames; no persistent user data yet	Local filesystem (uploads/) — planned: PostgreSQL/MongoDB